Frequentist validation of a gravitational-wave dispersion test: two pre-registered analyses stopped by their own coverage rules

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Abstract

We built a frequentist test for a quadratic-in-frequency modification of the gravitational-wave group velocity, parametrised by a length-squared Λ, and pre-registered it with stopping rules on frequentist coverage. The test works on residuals of public LIGO–Virgo–KAGRA strain against the published posterior samples, removes a posterior-derived subspace, treats calibration as a known covariance, and builds Neyman confidence sets by profile scans in Λ.

The primary analysis required two waveform models. Both pre-registered versions stopped before any on-source estimate. In version 1.0, injections of posterior samples into real off-source noise were covered by the 90% sets in 300 of 360 cases for the effective-one-body (EOB) models, below the pre-registered threshold of about 0.859; IMRPhenomXPHM passed with 313 of 352. A pre-specified diagnostic did not identify the mechanism. Version 2.0, whose belts include the posterior–reference mismatch, raised EOB coverage only to 299 of 352 and stopped again.

A separately registered, explicitly secondary IMRPhenomXPHM-only analysis gives 90% sets of $[-9.4, +5.2] \times 10^{-11}$ m² for 24 O1–O3 events and $[-3.3, +1.8] \times 10^{-11}$ m² for 20 O4a events, both containing $\Lambda = 0$ and compatible with the GWTC-4.0 bound. Two post-result checks show that these bounds are not validated: coverage at a true Λ of $\pm 2\sigma$ is 0.55, and an IMRPhenomXPHM–SEOBNRv5PHM waveform difference leaks into Λ (coverage 0.845). The main results are methodological: asymptotic thresholds under-cover strongly in the non-linear regime, and waveform-model differences, not noise, limit this class of test.

1 Introduction

A frequency-dependent group velocity of gravitational waves would accumulate a phase over cosmological distances that is absent in general relativity. Tests of this kind follow the modified-dispersion framework of Mirshekari, Yunes and Will [1] and are part of every LIGO–Virgo–KAGRA testing-GR release [2, 3, 6]. The most recent combined bound from GWTC-4.0, for the term quadratic in frequency in the group velocity, corresponds to $\Lambda \in [-2.4 \times 10^{-11}, +7.4 \times 10^{-12}]$ m² at 90% [3].

These bounds are Bayesian credible intervals. Their frequentist coverage, the fraction of repeated experiments in which the interval contains the true value, is rarely measured directly. Coverage matters in exactly the regime where dispersion tests are most sensitive: high frequencies, low masses and large phase shifts, where the likelihood is strongly non-linear in Λ and degenerate with the source parameters and with detector calibration [4].

This work had two aims. The first was an independent bound on Λ from public data. The second, which became the main result, was to build the test so that it could not report a number it had not earned: every stage was frozen in a public repository before the next stage was run, and stopping rules on coverage, fixed in advance, decided whether any on-source estimate was allowed.

The contributions are: (i) a residual-based, profile-likelihood construction with Neyman belts and a measured, not assumed, coverage; (ii) a complete audit trail in which two pre-registered versions of the primary analysis stopped themselves at their own off-source coverage rule because of the effective-one-body waveform labels; (iii) a pre-specified diagnostic of that failure; (iv) a separately registered secondary analysis with IMRPhenomXPHM; and (v) two post-result validation checks that locate the limits of the approach in waveform-model differences.

Section 2 defines the signal model and the data, Section 3 the method and the stopping rules, Section 4 the audit trail. Sections 5–7 report the two primary versions and the diagnostic, Section 8 the secondary analysis, Section 9 the validation checks, and Section 10 discusses what is and is not claimed.

2 Signal model and data

We use the group-velocity convention of GWTC-4.0 [3]. A dispersion term with frequency exponent $\alpha = 4$ adds to the frequency-domain phase

```
\delta\Psi(f) = -\frac{4\pi^{3}\,\Lambda\, I_{4}(z)}{c^{3}}\, f^{3} \equiv K\,\Lambda\, f^{3},
```

where $I_4(z)$ is the distance integral of that convention, evaluated at the posterior median redshift of each event, and Λ has units of m^2. The LVK amplitude A_4 in eV^{-2} converts as $\Lambda = A_4\,(\hbar c)^2 = A_4 \times 3.894 \times 10^{-14}\ \mathrm{m}^2$, so their 90% range $A_4 \in [-620, +190]\ \mathrm{eV}^{-2}$ becomes $[-2.4 \times 10^{-11}, +7.4 \times 10^{-12}]\ \mathrm{m}^2$. The particle-velocity form implemented in LALSimulation differs by a factor -3 and was not used; a convention check reproduces the GWTC-4.0 form to a ratio of 1.0000.

The f^2, f^3 and f^4 templates are correlated at about 0.97 after the subspace removal of Section 3. The test therefore constrains smooth dispersion of the class $\alpha \approx 2$–4, and a non-zero result could not be attributed to Λ alone.

Events. Two cohorts were fixed before any on-source computation:

Cohort	Data	PE labels	Events (after pair exclusions)	Waveform models
Search	O1–O3	GWTC-2.1/3, C01	24 (IMRPhenomXPHM), 25 (SEOBNRv4PHM)	IMRPhenomXPHM; SEOBNRv4PHM
Confirmation	O4a	GWTC-4.0, C00	20 and 20	IMRPhenomXPHM-SpinTaylor; SEOBNRv5PHM

Before the pair exclusions of Section 3 the search cohort had 27 events. Excluded by pre-registered rules were GW191204_171526 (no calibration envelope), GW200225_060421 (reference fit), GW231123_135430 (waveform systematics noted by LVK), three single-detector events, and four events whose labels lack recalibration priors (GW170608, GW190707_093326, GW190728_064510, GW190924_021846). All 25 SEOBNRv4PHM search labels come from LALInference; all other labels used come from bilby.

Strain. Public GWOSC strain, 134 files of 4096 s, with SHA-256 checksums recorded in the repository. Each event is analysed on the segment and frequency band of its PE configuration.

3 Method

The estimate of Λ comes from the part of the data that the published GR fit leaves unexplained, after every direction that a change of the source parameters could produce has been removed.

Residual. The reference waveform h_ref is generated from the stored maximum-likelihood posterior sample with the exact settings of its PE label, projected on each detector and multiplied by the mean calibration factor. A common time shift (±100 ms, 0.05 ms steps) and a complex amplitude are fitted, and the residual is r = d − a h_ref e^(−2πifΔt). All quantities are whitened by the PE noise spectra and stacked over detectors.

Nuisance subspace. From a fixed permutation of the posterior samples (seed 20261001), the first 800 usable samples form set A and the next 800 set B. The principal components of $h(\theta) - h_{ref}$ that carry 99% of the variance, together with the time, amplitude and phase directions, span the subspace Q, which is projected out. Calibration is not projected out; it enters as a covariance $C = I + JJ^T$, with J the first-order directions of the

bilby cubic-spline recalibration priors, and the estimator is generalised least squares. A synthetic test showed that projecting calibration out would absorb almost all of the f^3 signal.

Profile scan and Neyman construction. On a grid of ±8 times the smallest σ_lin of the cohort (step 0.05 σ_lin), the model $s(\Lambda) = P[h_{ref}(e^{iKf^3\Lambda} - 1)]$ is scanned with the exact phase. Per event and catalog, $q(t) = \chi^2(t) - \min \chi^2$ is compared with critical values c_t from 5000 belt trials of Gaussian noise plus calibration draws. The reported 90% set is the union of the sets obtained with subspaces A and B, so that the instability of the principal components widens the set instead of being ignored. Wilks′ threshold of 2.71 under-covers: per event coverage fell to 0.46, and catalog coverage to 0.80–0.86.

Stopping rules (fixed before any on-source estimate):

Rule	Condition	Action
0	Gaussian-belt coverage at five true values below the 99% binomial interval (≈ 0.883 for 2000 trials), or > 1% of intervals at the grid edge	stop
0a	off-source real-noise coverage of posterior-sample injections, per model, both cohorts pooled, below the 99% binomial interval (≈ 0.859)	stop before any on-source estimate
1	null-injection bias of a pair > 0.3 σ_lin	pair leaves the primary test
1a	median A–B difference of a pair > 0.5 σ_lin	pair leaves the primary test
2a	two-sided KS test of GR leakage, p < 0.01	pair leaves the primary test
3	calibration-only injections biased by > 1 σ_cat or excluding zero	no bound reported

Off-source stage (rule 0a). For each event–model pair, eight segments of real strain away from the event (guarded against every catalogued event) receive two injections: a posterior sample not used for the subspaces, and the reference waveform. Each passes through the full chain, refinement included, and q(0) is compared with the event′s own critical value. This stage tests non-Gaussian noise and the mismatch between signal and reference; it does not test calibration, since software injections bypass it.

4 Pre-registration and audit trail

Every decision was committed to the public repository before the computation it governs, and every result was committed before the next stage was run. The design document (v0.9.x, frozen as v1.0) fixed the cohorts, the estimator, the stopping rules and the order of stages: build, checks, catalog, off-source, and only then on-source, search cohort first.

Date (2026)	Tag or commit	Step
Oct 3	`v1.0-module1`	v1.0 code and design frozen (hash in FREEZE_v1.0.json)
Oct 6	`v1.1-report`	reporting layer frozen before its first use
Oct 6	5e03353	Addendum D: pair-level exclusions, per-model lists
Oct 7	`v1.0-stopped-0a`	rule 0a fired for EOB; v1.0 stops before any on-source estimate
Oct 7	d96415a	diagnostic of the failure specified before it was run
Oct 8	c274c7f	diagnostic report: no mechanism identified
Oct 8	5fa1401	v2.0 design draft (amended twice before freeze)
Oct 9	`v2.0-module1`	v2.0 frozen
Oct 11	68df69d	interim record: reading of rule 0a fixed before the last v2.0 result
Oct 11	`v2.0-stopped-0a`	rule 0a fired again for EOB; v2.0 stops
Oct 11	314b5d6	registration of the secondary IMRPhenomXPHM analysis
Oct 11	`supplement-xphm-frozen-r2`	secondary code frozen (r2 after a serialisation fix)
Oct 11	`supplement-search-result`	search on-source set committed before the confirmation run
Oct 12	`supplement-confirm-result`	confirmation on-source set committed
Oct 12	916c038, `supplement-validation-frozen`	post-result validation checks registered, code frozen

No threshold, cohort or construction choice was changed after a result was seen. Changes made before a freeze (two amendments of the v2.0 draft, one serialisation fix of the secondary code) are recorded with their reason in the corresponding commits.

5 Version 1.0: the primary analysis stops at rule 0a

Version 1.0 passed every synthetic check and then failed the real-noise coverage rule for the EOB models, so no on-source estimate was made.

Checks and exclusions. All 94 event–model pairs (47 events, two models) completed the stage-2 checks under the frozen code. The registered pair-level rules removed five pairs from the search cohort: IMRPhenomXPHM for GW190512_180714, GW190620_030421 and GW190915_235702, and SEOBNRv4PHM for GW170823 and GW191109_010717. No confirmation pair was removed.

Catalogs. The Neyman belts covered at 0.92–0.95 at all five true values (the union of A and B sets is conservative by construction), while Wilks′ threshold covered at 0.80–0.86. The expected 90% interval of the SEOBNRv4PHM search catalog was $[-3.7, +3.8] \times 10^{-11}$ m². Information is concentrated in a few events: the effective number of contributing events was about 3, and removing GW191129_134029 widened the expected search interval by about 30%.

Rule 0a. Coverage of posterior-sample injections in real off-source noise:

Model	Search	Confirmation	Pooled	99% lower end	Rule 0a
IMRPhenomXPHM	167/192 = 0.870	146/160 = 0.912	313/352 = 0.889	0.859	pass
EOB (SEOBNRv4PHM / v5PHM)	155/200 = 0.775	145/160 = 0.906	300/360 = 0.833	0.859	**stop**

The exact binomial probability of 300 or fewer covered out of 360 at a true coverage of 0.90 is below 10^{-3}. With the reference waveform itself injected, search SEOBNRv4PHM coverage was 170/200 = 0.850, so the deficit is larger for injected posterior samples than for the reference. Per event, four search events reached only 4/8 (GW150914, GW170814, GW190706_222641, GW200129_065458); under 0.90 coverage, four such events among 25 would be expected about 0.1 times.

As registered, the analysis stopped before any on-source estimate (tag `v1.0-stopped-0a`).

6 Diagnosis of the v1.0 failure

A diagnostic specified before it was run (events, variants, metric and threshold fixed in advance) did not identify the mechanism. It used only the existing off-source segments and could not change the v1.0 result.

The four failing SEOBNRv4PHM events were compared with four controls (the search events with the smallest reference-time refinement) and with the IMRPhenomXPHM labels of the same four events. A first, exact gate required the recomputed v1.0 flags to reproduce the stored ones; all 12 pairs reproduced with zero mismatches.

Measurement 1: time offsets. Aligning 200 posterior samples per label to the reference showed that SEOBNRv4PHM samples scatter by 2–18 ms, in failing and control events alike, while IMRPhenomXPHM samples scatter by 0.1–0.6 ms. Large offsets occurred in a failing event (GW190706_222641, median −22.5 ms) and in a well-covered control (GW200112_155838, +37.5 ms). Time misalignment alone does not separate failure from success.

Gate 2: two variants on the same segments, injections, belts and seeds.

Variant	Failing EOB events	Controls	Failing events with IMRPhenomXPHM
v1.0	16/32	27/32	30/32
(a) reference at the posterior medoid	20/32	28/32	29/32
(b) samples aligned in time and phase	23/32	30/32	26/32

The pre-registered threshold was 24/32, the lower end of the 99% interval of Binomial(32, 0.90); neither hypothesis was supported. The stored reference lay 13–40 whitened noise units from the posterior medoid in the failing events, but re-centring did not restore coverage consistently (GW190706_222641 rose from 4/8 to 7/8, GW200129_065458 fell from 4/8 to 2/8), and alignment lowered the IMRPhenomXPHM cross-check from 30/32 to 26/32.

Sampler audit (descriptive, after the result). All 25 search SEOBNRv4PHM labels come from LALInference; the SEOBNRv5PHM and IMRPhenomXPHM labels come from bilby. Waveform model and sampler are therefore fully confounded, and the public O1–O3 release contains no other EOB label that could separate them.

7 Version 2.0: mismatch-including belts, and a second stop

Version 2.0 changed one thing and kept both models: the belts no longer assume that the true signal equals the reference. It improved the failing cohort but stopped again at rule 0a.

The change. Every belt and check trial adds, to Gaussian noise and calibration draws, the projected residual that the analysis chain leaves of a GR signal at a posterior sample θ not used for that subspace:

```
n_{\mathrm{trial}} = n_{\mathrm{G}} + n_{\mathrm{cal}} + P_S\big[\, h(\theta)
- a\, h_{\mathrm{ref}}\, e^{-2\pi i f \Delta t} \big],
```

with (Δt, a) fitted noise-free exactly as in the chain. The belts of subspace A use 400 samples of set B, and vice versa. Geometry, cohorts, models, rules and thresholds are those of v1.0.

Two amendments before the freeze, both found by the synthetic test and recorded with their reason: a separate sample set was impossible because the LALInference labels have about 1600–4000 samples (GW150914: 1642), hence the cross-set rule; and the raw difference h(θ) − h_ref had an rms of 39.7 noise units for GW150914 SEOBNRv4PHM, almost entirely the 2–18 ms time scatter that the chain removes, against 4.9 for the refined residual (3.2 for GW170104 IMRPhenomXPHM). Waveforms were generated in isolated worker processes because native crashes of the EOB code had silently ended a run.

Catalogs. Rule 0 passed in all four catalogs (lowest Neyman coverage 0.919). The expected 90% interval of the SEOBNRv4PHM search catalog widened from [−3.7, +3.8] to [−4.6, +4.6] × 10^{-11} m², about 20–25%.

Rule 0a on new data. Validation used new off-source segments (none overlapping v1.0) and a new injection set. The reading of the pooled rule, and the commitment to report the search shortfall separately, were committed before the last cohort's result was known.

Model	Search	Confirmation	Pooled	99% lower end	Rule 0a
IMRPhenomXPHM	171/192 = 0.891	135/152 = 0.888	306/344 = 0.890	0.858	pass
EOB	167/200 = 0.835	132/152 = 0.868	299/352 = 0.849	0.859	**stop**

Search SEOBNRv4PHM coverage rose from 0.775 to 0.835 and reference-waveform coverage from 0.850 to 0.915, but the pooled EOB coverage stayed below the threshold (exact binomial probability about 10^{-3}). The mismatch term explains part of the v1.0 failure, not all of it. v2.0 stopped before any on-source estimate (tag `v2.0-stopped-0a`).

8 Secondary analysis with IMRPhenomXPHM

Both on-source sets contain Λ = 0 and are compatible with the GWTC-4.0 bound; they come from an analysis selected after the EOB failures and are reported as secondary.

Registration. Before any on-source quantity existed, and before the code that computes it was written, a separate registration fixed: IMRPhenomXPHM only; the v2.0 construction as frozen (v1.0 not computed, to leave no choice between constructions); the Addendum D pairs; all stopping rules for this model alone; search cohort first, committed before the confirmation run; no combination of cohorts; and the reading of a set that excludes zero (a candidate of the class $\alpha \approx 2$–4, never evidence for Λ). The on-source chain, applied to off-source segments, reproduced the stored v2.0 q(0) and c(0) exactly before the code was frozen.

Gate. Rule 0 passed (lowest Neyman coverage 0.919 and 0.920), rule 0a passed (306/344 = 0.890), and rule 3 passed in both cohorts: 100 noise-free calibration-only injections gave a median estimate of zero and no set excluding zero. The on-source refinement was small for every event (time shift within ±0.35 ms, |a| between 0.98 and 1.11).

Results.

Cohort	Events	90% set (10^{-11} m²)	Contains 0	Without the most informative event (10^{-11} m²)
Search, O1–O3	24	[−9.44, +5.21]	yes	[−6.10, +10.5] without GW191129_134029
Confirmation, O4a	20	[−3.27, +1.75]	yes	[−3.51, +2.03] without GW231224_024321
LVK GWTC-4.0 [3]	83	[−2.4, +0.74]	yes	—

Both sets are contiguous and away from the grid edge. The confirmation set is about 1.6 times wider than the LVK interval, the search set about 4.7 times.

Influence of single events (descriptive, decided after the results). Removing any single event leaves $\Lambda = 0$ inside the set in both cohorts. Information is concentrated in three events: GW191129_134029, GW191216_213338 and GW190412 carry 82% of the search weight, and GW231224_024321, GW230627_015337 and GW231020_142947 carry 93% of the confirmation weight. Removing one of them widens the set by 10–30%; removing any other event changes it by at most one or two grid steps.

9 Post-result validation: the bounds are not validated

Two checks, registered after the on-source results and run only on new off-source segments with new posterior samples, both fell below the rule-0a threshold. They cannot change the sets of Section 8; they say how far those sets can be trusted.

Both checks use IMRPhenomXPHM, the frozen chain and belts, eight new segments per pair (ladder index k ⩾ 16, none used before) and eight new posterior samples per pair, with the acceptance threshold of rule 0a.

Check	Injection	Covered	99% lower end	Estimate minus truth (σ_lin), median and 16–84%
V1	posterior sample with dispersion at Λ = ±2 σ_lin of the event	189/344 = 0.549	0.858	−0.55 and +0.65 for +2σ and −2σ; tails to ∓3
V2	SEOBNRv5PHM waveform with the same parameters, Λ = 0	277/328 = 0.845	0.857	+0.00; [−1.98, +1.98]

V1: coverage away from zero. The two signs behave symmetrically, which excludes a sign error. Estimates are pulled toward zero, by about 0.6 σ_lin at the median and by more than 2 σ_lin in about one case in six. The effect grows with σ_lin. The four most informative events reach 6, 6, 5 and 7 of 8 with almost no pull (24/32 = 0.75), while weaker events, for which ±2 σ_lin is a large Λ and a dispersion phase of many radians, show pulls of 1 to 4.8 σ_lin and coverage of 1–3 of 8. A likely cause, not yet tested, is that the time and phase refinement of the chain absorbs part of a large dispersion phase, which the belts do not model; at Λ = 0 this has no effect, which is why rule 0a passed.

V2: waveform-model leakage. With signals from SEOBNRv5PHM analysed by the IMRPhenomXPHM chain, coverage at Λ = 0 falls from 0.890 to 0.845. The estimates are unbiased but have extra scatter comparable to the statistical width. GW230627_015337 and GW230924_124453 were skipped as registered, because the (4,4) ringdown of SEOBNRv5PHM exceeds the Nyquist frequency of their label settings.

Consequence. The sets of Section 8 remain consistent with Λ = 0, but their end points are not validated: they are likely too narrow and they depend on the waveform model. They should be read as indicative.

10 Discussion

No dispersion is detected, no validated bound is claimed, and the method's own rules located its weak point: waveform-model differences.

What is claimed. Public data analysed with IMRPhenomXPHM give sets consistent with Λ = 0 in two independent cohorts, compatible with GWTC-4.0. The pre-registered stopping rules worked as intended: twice they stopped an analysis whose intervals were too

narrow in real noise, before any on-source number existed. **What is not claimed:** a validated bound (Section 9), a two-model result (Sections 5 and 7), or a test of Λ specifically rather than of the $\alpha \approx 2$–4 class (Section 2).

Asymptotic thresholds. Wilks' threshold under-covered by up to a factor of two per event in the non-linear regime (0.46 instead of 0.90). Any profile-likelihood bound on dispersion at high frequency and low mass needs a Neyman construction, or an equivalent coverage measurement. This does not concern Bayesian credible intervals directly, which do not promise frequentist coverage; whether published dispersion intervals have it is an open question that injection campaigns could settle.

Waveform models are the limiting systematic. Three results point the same way: the EOB failures of rule 0a in both versions, the leakage of an IMRPhenomXPHM–SEOBNRv5PHM difference into Λ (V2), and the absence of any noise- or calibration-driven failure (rules 0 and 3 passed). The cause of the EOB failure is not identified; model and sampler are confounded in the public release (Section 6). Calibration with realistic priors had a negligible effect, consistent with recent work on calibration misspecification in dispersion tests [4].

A known inconsistency. The chain refines time and phase before scanning, but the belts do not model that refinement. At $\Lambda = 0$ this is harmless; at large Λ it probably absorbs part of the signal (V1). A next version should include the refinement inside every belt trial, or fit it jointly with Λ, and should marginalise the belts over waveform models.

Limitations. The secondary analysis uses one waveform model and was selected after the EOB failures. Information is concentrated in about three events per cohort. Four low-mass events among the most sensitive to dispersion (GW190924_021846 alone has $\sigma_{\mathrm{lin}} \approx 2.4 \times 10^{-11}$ m^2 with SEOBNRv4PHM) were excluded by the calibration rule; their pre-specified diagnostic was not executed because its condition, a primary result, did not occur. The O1–O3 calibration priors, from which J and the mean calibration factor were built, carry a sign error in the means of the LIGO priors [5]; the means are small compared with the widths and rule 3 found no calibration-driven bias, but the effect on J was not quantified separately.

Outlook. O4b data, analysed blind with a frozen and validated construction, would test these conclusions on independent events. A model-specific prediction, for example a required sign of Λ, registered before such data are opened, is what would turn a class-level bound into a test of a particular theory.

Data and code availability

All code, design documents, registrations, logs and results are public at github.com/Kretski/residual-first-framework (directory `scanner/gw`). Each stage can be traced to the tags of Section 4; the freeze manifests record the SHA-256 hashes of the code that produced every result. An archived release with a DOI will accompany submission.

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Still to add before submission: the software and waveform references (LALSuite, bilby, pyseobnr, IMRPhenomXPHM, SEOBNRv4PHM, SEOBNRv5PHM, LALInference), the GWOSC data papers, and Neyman (1937), Wilks (1938), Feldman and Cousins (1998).